Topic: UNSDG Goal 11 Make cities and human settlements inclusive, safe, resilient and sustainable . Subtopic: *Building Sensory-Friendly Urban Futures*

UNGoal: Goal 11 | Department of Economic and Social Affairs

SDG 11 focuses heavily on making urban environments accessible to everyone, with a specific emphasis on vulnerable populations, including people with disabilities.

Target 11.2 (Accessible Transit): Provide access to safe, affordable, accessible, and sustainable transport systems for all, improving road safety, notably by expanding public transport, with special attention to the needs of those in vulnerable situations, women, children, persons with disabilities, and older persons.

Target 11.7 (Inclusive Public Spaces): Provide universal access to safe, inclusive and accessible, green and public spaces, in particular for women and children, older persons and persons with disabilities.

New articles/ Research

- Metro Trains in Melbourne launch the Hidden Disabilities Sunflower
- Designing cities for neurodiversity – why we need inclusive public open spaces | City Futures Research Centre Blog
- Designing for neurodivergence makes life better according to new research | Architecture & Design

Problem Statement

Neurodivergent sensory-sensitive individuals represent **young adults** in Australia, a demographic that exhibits a reliance on public transportation for daily mobility. Despite this dependency, these individuals currently **lack a dedicated, real-time, sensory-aware wayfinding infrastructure to navigate the dense urban environment whilst travelling to work in the City of Melbourne.** ***Within these high-density corridors including active construction, public demonstrations and heightened law enforcement operation*** - introduce severe sensory stressors.While accessible, sensory-friendly public spaces are critical, navigating these environments independently remains a significant challenge

(How) How might we transform Melbourne's CBD into a naturally navigable environment for sensory sensitive adults?

Epics and User Stories, Data sources, DoD, description and Benefits

Epic 1: Sensory-Aware Route Planning & Real-Time Navigation

As a sensory-sensitive commuter, I want to identify and follow the least sensory-intensive route to my destination, so that I can travel independently with less anxiety and sensory overload.

Description Enable neurodivergent and sensory-sensitive adults to independently navigate Melbourne CBD using routes optimized for lower sensory load rather than shortest travel time.

Benefits:

- Reduce exposure to high-density pedestrian areas.
- Increase confidence in independent travel.
- Improve route completion rates without sensory distress.
- Increase adoption of sensory-friendly route recommendations.

User Story 1.1 **As a sensory-sensitive commuter,** I want to see a sensory indicator (High or Low) for different routes, so that I can choose the path that feels safest and least overwhelming.

Data Sources

- Pedestrian Counts per Hour [data.gov.au]
- Sensor Locations [data.gov.au]

User Story 1.2 **As a sensory-sensitive commuter,** I want to avoid highly congested pedestrian corridors, **so that** I can reduce exposure to crowd-related stress.

Data Sources

- Real-time Pedestrian Counting System [melbourne.vic.gov.au], [data.gov.au]

User Story 1.3 **As a sensory-sensitive commuter during peak hour, I want** alternative low-stimulation routes when crowd density exceeds my preferred threshold, **so that** I can avoid unexpected sensory overload.

Data Sources

- Pedestrian Counts per Hour [data.gov.au]

Definition of Done

✅ User can enter a destination within Melbourne CBD.

✅ System generates at least one sensory-aware route using City of Melbourne Open Data sources, including pedestrian density information. [melbourne.vic.gov.au], [data.gov.au]

✅ Routes are assigned a sensory indicator based on agreed factors (e.g., pedestrian volume, construction activity, events).

✅ Route recommendations dynamically adjust when crowd levels exceed user-defined limits.

✅ Walking routes integrate with public transport access points.

✅ Accessibility and usability testing completed with representative or the closest neurodivergent users.

✅ All identified critical and high-priority defects are resolved.

✅ Acceptance criteria are met and approved by mentors

Epic 2.0 Sensory Environment Monitoring

As a sensory-sensitive commuter ,I want to receive real-time alerts about sensory stressors and nearby quiet spaces, so that I can proactively avoid overwhelming environments and maintain my wellbeing while travelling.

Description :

Provide real-time visibility of sensory stressors across the CBD and enable users to make informed navigation decisions before entering overwhelming environments.

Benefits:

- Reduce unexpected exposure to sensory triggers.
- Increase successful rerouting before reaching high-stimulation zones.
- Improve user perception of safety and comfort.
- Increase usage of identified sensory refuge locations.

US 2.1 As a sensory-sensitive commuter, I want to identify nearby sensory refuge locations such as parks, libraries and quiet public spaces **so that** I can take breaks when feeling overwhelmed.

Data Sources

- Public facilities and open space datasets [melbourne.vic.gov.au]

US 2.2 As a sensory-sensitive commuter, I want predictive alerts showing which areas are likely to become overwhelming within the next hour, **so that I can adjust my travel plans proactively.**

Data Sources

- Historical pedestrian patterns and hourly trends [data.gov.au], [melbourne.vic.gov.au]

EPIC 2.0 Definition of Done

✅ Users receive real-time alerts for high-density pedestrian areas. [melbourne.vic.gov.au], [data.gov.au]

✅ Nearby sensory refuge locations (parks, libraries, quiet public spaces) are displayed on demand. [melbourne.vic.gov.au]

✅ Predictive alerts are generated using historical pedestrian trends. [data.gov.au], [melbourne.vic.gov.au]

✅ Alert accuracy has been validated against available city data.

✅ All critical and high-priority defects are resolved.

✅ Accessibility testing completed and approved.

✅Aacceptance criteria are met and approved by mentors